

# A Two-State Counterexample to Erbar–Fathi Conjecture 6.9

Counterexample packet for arXiv:1612.00514

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**Status: candidate full counterexample, likely valid.** The intended assertion that exponential concentration with rate  $\rho$  and nonnegative entropic Ricci curvature force a spectral gap  $\lambda \geq C(M)\rho^2$  fails on imbalanced two-state chains. The dimensionally inconsistent  $\rho^{-2}$  exponent printed in the source is separately false by time rescaling.

## 1 Source and exact conjecture

The source is M. Erbar and M. Fathi, *Poincaré, modified logarithmic Sobolev and isoperimetric inequalities for Markov chains with non-negative Ricci curvature*, arXiv:1612.00514, later JFA 274 (2018), 3056–3089 [1]. For a reversible continuous-time chain  $(\mathcal{X}, Q, \pi)$ , the source writes  $P(\lambda)$  for

$$\mathrm{Var}_\pi f \leq \lambda^{-1} \mathcal{E}(f, f).$$

Its entropic transport distance  $\mathcal{W}$  induces  $d_{\mathcal{W}}(x, y) = \mathcal{W}(\delta_x, \delta_y)$ . Conjecture 6.9 on source page 23 assumes  $\mathrm{Ric}(\mathcal{X}, Q, \pi) \geq 0$  and concentration

$$\pi(A_r^c) \leq M e^{-\rho r} \quad \text{for every } A \text{ with } \pi(A) \geq \tfrac{1}{2},$$

and asks for a Poincaré bound depending only on  $M$  and  $\rho$ .

The displayed formula in [1] is  $P(C(M)\rho^{-2})$ . Since the Poincaré parameter has units of inverse distance squared, the intended formula is  $P(C(M)\rho^2)$ . This is also how Münch explicitly restates Conjecture 6.9, as  $\lambda \geq C(M)\beta^2$ , in his later Ollivier-curvature analogue [2, §6.1]. We disprove the intended formula nontrivially and the literal printed formula by an independent scaling argument.

## 2 Proof intuition

Put almost all stationary mass at one endpoint of a two-state chain. Below the single nonzero metric scale, the concentration tail is just the mass  $p$  of the rare endpoint. It can therefore be written as  $e^{-\rho r}$  with  $\rho D = \log(1/p)$ .

The intrinsic entropic distance does react to the small atom, but only as  $D^2 = O(\log(1/p))$ . Consequently  $\rho^2$  grows like  $\log(1/p)$ , while the generator is normalized to have spectral gap one. An exact one-dimensional calculation also shows strictly positive entropic Ricci curvature. Thus atomic concentration at sub-mesh scales is too strong to quantitatively control the spectral gap.

## 3 The two-state geometry

Fix  $0 < p \leq 1/4$ , put  $q = 1 - p$ , and consider

$$\mathcal{X} = \{0, 1\}, \quad Q(0, 1) = p, \quad Q(1, 0) = q, \quad \pi(0) = q, \quad \pi(1) = p.$$

profiles.

As we have seen in the previous sections, the alternative approach of Gozlan, Roberto and Samson [15], based on functional inequalities, is more easily adapted to the discrete setting. While unlike Milman, they do not recover the Gaussian isoperimetric inequality, they nonetheless show that when curvature is bounded from below, a strong enough Gaussian concentration implies a logarithmic Sobolev inequality. However, we have not been able to adapt a key step in their approach, which is that Gaussian concentration implies a weak transport-entropy inequality. In the discrete setting, the analogous inequality we would need would be

$$\mathcal{W}(\mu, \pi)^2 \leq c_1 \text{Ent}_\pi(\mu) + c_2.$$

To establish it, we would need to better understand the relationship between bounds on  $\mathcal{W}$  and concentration. An important difference between the discrete and the continuous situation is that lack of a dual Kantorovich formulation for the distance  $\mathcal{W}$ .

Nonetheless, we state as conjectures the discrete analogues of the results of [32, 31, 33, 22, 15]:

**Conjecture 6.9.** *Assume that  $\text{Ric}(\mathcal{X}, Q, \pi) \geq 0$  and that the invariant measure  $\pi$  satisfies a concentration property w.r.t. the distance  $d_{\mathcal{W}}$  with profile  $\alpha(r) = Me^{-\rho r}$ . Then there exists a constant  $C(M)$  such that  $PI(C(M)\rho^{-2})$  holds.*

**Conjecture 6.10.** *Assume that  $\text{Ric}(\mathcal{X}, Q, \pi) \geq -\kappa$  for some  $\kappa > 0$ , and that a concentration property with respect to the distance  $d_{\mathcal{W}}$  holds with profile  $\alpha(r) = Me^{-\rho r^2}$ . Then there exists a constant  $\tau(M)$  and  $\lambda(\kappa, M, \rho)$  such that if  $\frac{\kappa}{\rho} < \tau(M)$  then  $MLSI(\lambda(\kappa, M, \rho))$  holds. If moreover  $\text{Ric}(\mathcal{X}, Q, \pi) \geq 0$  then  $MLSI(cM\rho)$  holds for some universal constant  $c$ .*

In the Riemannian setting, these results hold with no dependence on  $M$ , but for non-smooth geodesic spaces the proof of [15] has an extra dependence on  $M$  of the form we use in the statements of these conjectures.

As in the continuous setting, Theorem 6.5 already tells us that under these assumptions a mLSI holds. The open problem in Conjecture 6.10 is the value of the constant.

Figure 1: Conjectures 6.9–6.10 on page 23 of the source.

Detailed balance holds, and the nonzero eigenvalue of  $-L$  is  $p + q = 1$ . Hence the optimal Poincaré parameter is

$$\lambda_p = 1. \tag{1}$$

Parameterize a probability measure by  $\mu_m = (1 - m, m)$ ,  $0 \leq m \leq 1$ . Its density relative to  $\pi$  is

$$\rho_m(0) = \frac{1 - m}{q}, \quad \rho_m(1) = \frac{m}{p}.$$

Let  $\theta(a, b) = (a - b)/(\log a - \log b)$  be the logarithmic mean and put

$$\ell_p(m) = \log \frac{mq}{p(1 - m)}, \quad K_p(m) = pq \theta\left(\frac{1 - m}{q}, \frac{m}{p}\right) = \frac{m - p}{\ell_p(m)}.$$

The two directed edge terms and the factor  $1/2$  in the action give  $\mathcal{A}(\rho_m, \psi) = K_p(m)(\psi(1) - \psi(0))^2$ . The continuity equation gives  $\dot{m} = K_p(m)(\psi(1) - \psi(0))$  up to an irrelevant sign. Therefore the Riemannian line element and endpoint distance are

$$ds^2 = \frac{dm^2}{K_p(m)}, \quad D_p = d_{\mathcal{W}}(0, 1) = \int_0^1 \sqrt{\frac{\ell_p(m)}{m - p}} dm. \tag{2}$$

The quotient is positive, extends continuously at  $m = p$ , and has integrable endpoint singularities.

**Lemma 1** (Positive entropic curvature). *The chain satisfies  $\text{Ric}(\mathcal{X}, Q, \pi) \geq 1/2$ .*

*Proof.* The entropy along the line is

$$H_p(m) = m \log \frac{m}{p} + (1-m) \log \frac{1-m}{q}, \quad H'_p(m) = \ell_p(m), \quad H''_p(m) = \frac{1}{m(1-m)}.$$

For the one-dimensional metric tensor  $G = 1/K_p$ , the Riemannian Hessian relative to the metric equals

$$\frac{\text{Hess}_G H_p}{G} = K_p \left( H''_p - \frac{G'}{2G} H'_p \right) = \frac{1}{2} \left( 1 + \frac{K_p(m)}{m(1-m)} \right) \geq \frac{1}{2}.$$

This is precisely  $1/2$ -convexity of entropy along  $\mathcal{W}$ -geodesics in the interior. Continuity of the entropy and finiteness of (2) extend it to endpoint measures.  $\square$

**Lemma 2** (Logarithmic diameter bound). *For  $0 < p \leq 1/4$  and  $L = \log(1/p)$ ,*

$$D_p \leq 7\sqrt{L}, \quad \text{hence} \quad D_p^2 \leq 49L.$$

*Proof.* Write  $g(u) = \log(u/(1-u))$ , so the integrand in (2) is  $\sqrt{(g(m) - g(p))/(m-p)}$ . Split the integral at  $p, 2p, 1/2$ . On  $(0, p)$ , since  $(1-s)^{-1} \leq 4/3$  for  $0 < s \leq p$ ,

$$\int_0^p \sqrt{\frac{g(p) - g(m)}{p-m}} dm \leq \sqrt{\frac{4p}{3}} \int_0^1 \sqrt{\frac{\log(1/u)}{1-u}} du < \frac{4}{\sqrt{3}} \sqrt{p}.$$

The last universal integral is less than 2 (split at  $u = 1/2$ ). On  $(p, 2p)$ , the mean-value formula and  $g'(u) = 1/[u(1-u)]$  give an integrand at most  $\sqrt{2/p}$ , so this piece is at most  $\sqrt{2p}$ .

For  $2p \leq m \leq 1/2$ ,  $g(m) - g(p) \leq L$  and  $m - p \geq m/2$ , whence this piece is at most  $2\sqrt{L}$ . Finally, for  $1/2 \leq m < 1$ , we have  $m - p \geq 1/4$  and

$$g(m) - g(p) \leq L + \log \frac{1}{1-m}.$$

Using  $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$ , this last piece is at most

$$\sqrt{L} + 2 \int_0^{1/2} \sqrt{\log(1/u)} du \leq \sqrt{L} + \sqrt{\pi}.$$

Since  $L \geq \log 4$ , the sum of the four bounds is less than  $7\sqrt{L}$ .  $\square$

## 4 Counterexample to the intended conjecture

**Theorem 1.** *There is no positive constant  $C(1)$  such that every finite reversible chain with nonnegative entropic Ricci curvature and exponential concentration profile  $e^{-\rho r}$  satisfies  $P(C(1)\rho^2)$ . In particular, the intended form of Conjecture 6.9 is false.*

*Proof.* For the two-state chain, every set of  $\pi$ -mass at least  $1/2$  is either  $\{0\}$  or the whole space. For  $A = \{0\}$ ,

$$\pi(A_r^c) = \begin{cases} p, & 0 \leq r < D_p, \\ 0, & r \geq D_p. \end{cases}$$

Thus, with

$$\rho_p = \frac{\log(1/p)}{D_p},$$

the measure has concentration profile  $e^{-\rho_p r}$  (with  $M = 1$ ). Lemmas 1 and 2 give the curvature hypothesis and

$$\rho_p^2 = \frac{L^2}{D_p^2} \geq \frac{L}{49} \rightarrow \infty.$$

But (1) says that  $P(\lambda)$  holds only for  $\lambda \leq 1$ . The conjectured  $P(C(1)\rho_p^2)$  is therefore impossible for all small  $p$ .  $\square$

## 5 The literal printed exponent

**Theorem 2.** *The formula  $P(C(M)\rho^{-2})$  printed in Conjecture 6.9 is also false for every positive  $C(1)$ .*

*Proof.* Fix one  $p \in (0, 1/2)$  and multiply both transition rates by  $s > 0$ . The curvature remains nonnegative, the spectral gap becomes  $\lambda_s = s$ , and the entropic action is multiplied by  $s$ , so  $D_s = D_p/\sqrt{s}$ . The same exact concentration argument gives

$$\rho_s = \frac{\log(1/p)}{D_s} = \sqrt{s} \frac{\log(1/p)}{D_p}.$$

The literal assertion would require

$$s = \lambda_s \geq C(1)\rho_s^{-2} = \frac{C(1)D_p^2}{s \log^2(1/p)},$$

which fails as  $s \downarrow 0$ .  $\square$

## 6 Verification, novelty, and limitations

**Verification.** The proof uses only the exact two-state action and elementary integral bounds. The included regression script independently evaluates  $D_p$  by quadrature, checks the pointwise curvature formula on dense grids, verifies  $D_p^2 < 49 \log(1/p)$  down to  $p = 10^{-8}$ , and checks the rescaling law. It is not used as proof. The strict definition  $A_r^c = \{d(x, A) > r\}$  causes no endpoint issue: the tail is zero at  $r = D_p$ .

**Bounded novelty search.** The four run indexes were searched for arXiv:1612.00514, Conjecture 6.9, entropic curvature, exponential concentration, spectral gap, and two-state chains. Exact-phrase, title/citation, author, two-point, and close-variant web/arXiv searches through 17 August 2026 found no counterexample to the original entropic-curvature statement. Münch [2] explicitly claims affirmative analogues under nonnegative Ollivier curvature, but his results use a different curvature/metric framework and discrete-scale estimates; they do not establish the source's entropic- $d_W$  conjecture. Novelty is plausible, not certified.

**Scope and review.** This completely disproves Conjecture 6.9 under either exponent reading. It does not disprove or prove Conjecture 6.10; the same two-state family does not produce an MLSI contradiction. The mechanism exploits an atom below the first nonzero metric scale, suggesting that a repaired theorem must truncate the concentration rate at that scale or impose a non-atomic/geodesic regularity condition. Human review should focus on the normalization in (2), the one-dimensional Hessian calculation, and the source exponent interpretation.

## References

- [1] M. Erbar and M. Fathi, *Poincaré, modified logarithmic Sobolev and isoperimetric inequalities for Markov chains with non-negative Ricci curvature*, J. Funct. Anal. **274** (2018), 3056–3089; [arXiv:1612.00514](#).
- [2] F. Münch, *Ollivier curvature, isoperimetry, concentration, and log-Sobolev inequality*, [arXiv:2309.06493](#), 2023.